

```
In [1]: # Colab + Local compatible Loader (downloads CSV if not present)
import os
import pandas as pd
import numpy as np

CANDIDATES = ['Salary_dataset.csv', 'lab04/Salary_dataset.csv', '/content/Salary
path = next((x for x in CANDIDATES if os.path.exists(x)), None)
if path is None:
    url = 'https://raw.githubusercontent.com/manaal6/Introduction_To_AI_Lab_Task
    print('Downloading:', url)
    path = 'Salary_dataset.csv'
    pd.read_csv(url).to_csv(path, index=False)
print('Using:', path)
df = pd.read_csv(path)
df.head()
```

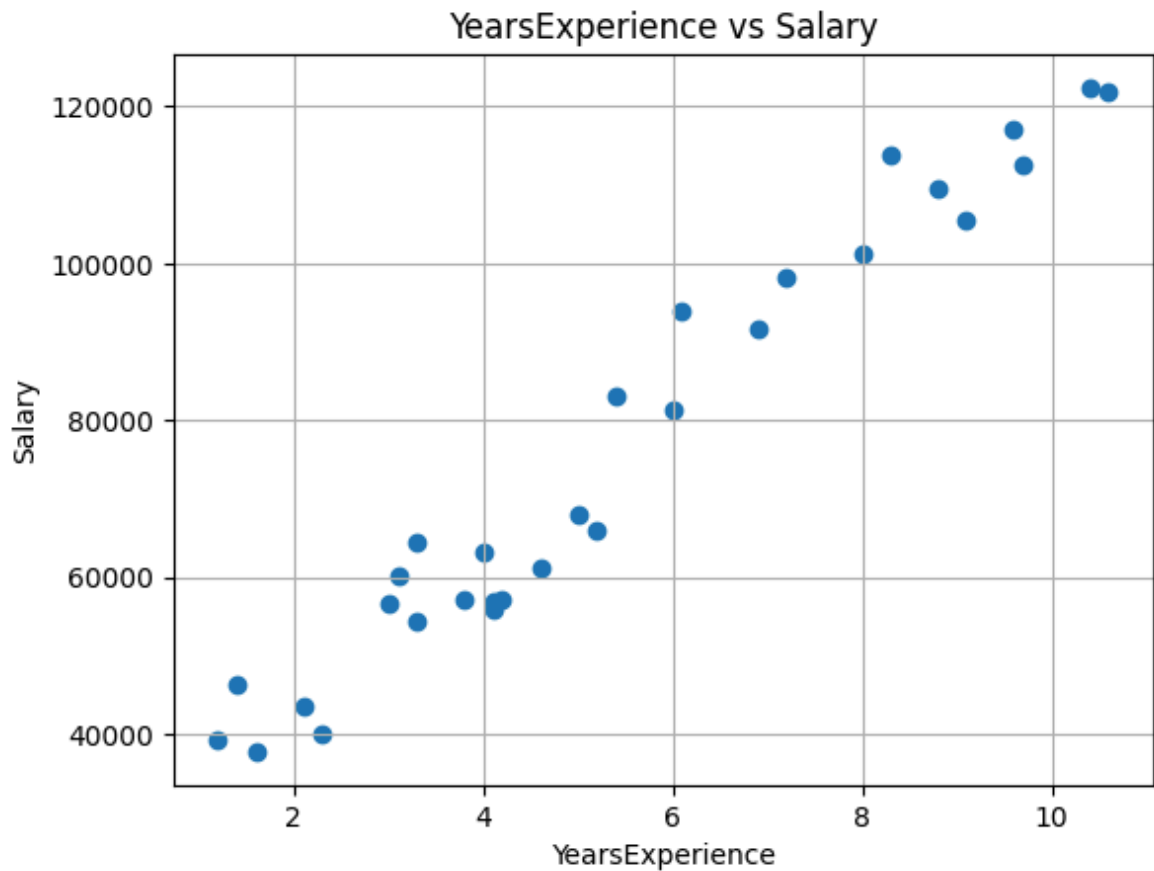
Using: Salary_dataset.csv

```
Out[1]:
```

	YearsExperience	Salary
0	1.2	39344
1	1.4	46206
2	1.6	37732
3	2.1	43526
4	2.3	39892

```
In [2]: import matplotlib.pyplot as plt
plt.scatter(
    df["YearsExperience"],
    df["Salary"]
)
plt.xlabel("YearsExperience")
plt.ylabel("Salary")
plt.grid("true")
plt.title("YearsExperience vs Salary")

plt.show()
```



```
In [3]: from sklearn.model_selection import train_test_split
X=df[['YearsExperience']]
y=df['Salary']
X_train, X_test, y_train, y_test = train_test_split(X,y,test_size=0.2,random_sta
```

```
In [4]: X_test
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```
Out[4]:
```

	YearsExperience
27	9.7
15	5.0
23	8.3
17	5.4
8	3.3
9	3.8

```
In [5]: from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

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In [6]: model = LinearRegression()
model.fit(X_train, y_train)
```

Out[6]:

▼ LinearRegression ⓘ ?

- Parameters
- Fitted attributes

```
In [7]: y_predict = model.predict(X_test)
y_predict
```

```
Out[7]: array([115791.21011287,  71499.27809463, 102597.86866063,  75268.80422384,
        55478.79204548,  60190.69970699])
```

```
In [8]: results = pd.DataFrame({
    'YearsExperience': X_test['YearsExperience'].values,
    'Actual Salary': y_test.values,
    'Predicted Salary': y_predict
})
print(results)
```

	YearsExperience	Actual Salary	Predicted Salary
0	9.7	112636	115791.210113
1	5.0	67939	71499.278095
2	8.3	113813	102597.868661
3	5.4	83089	75268.804224
4	3.3	64446	55478.792045
5	3.8	57190	60190.699707

```
In [9]: plt.figure(figsize=(8, 5))

plt.scatter(
    X_test["YearsExperience"],
    y_test,
    label="Actual Values"
)

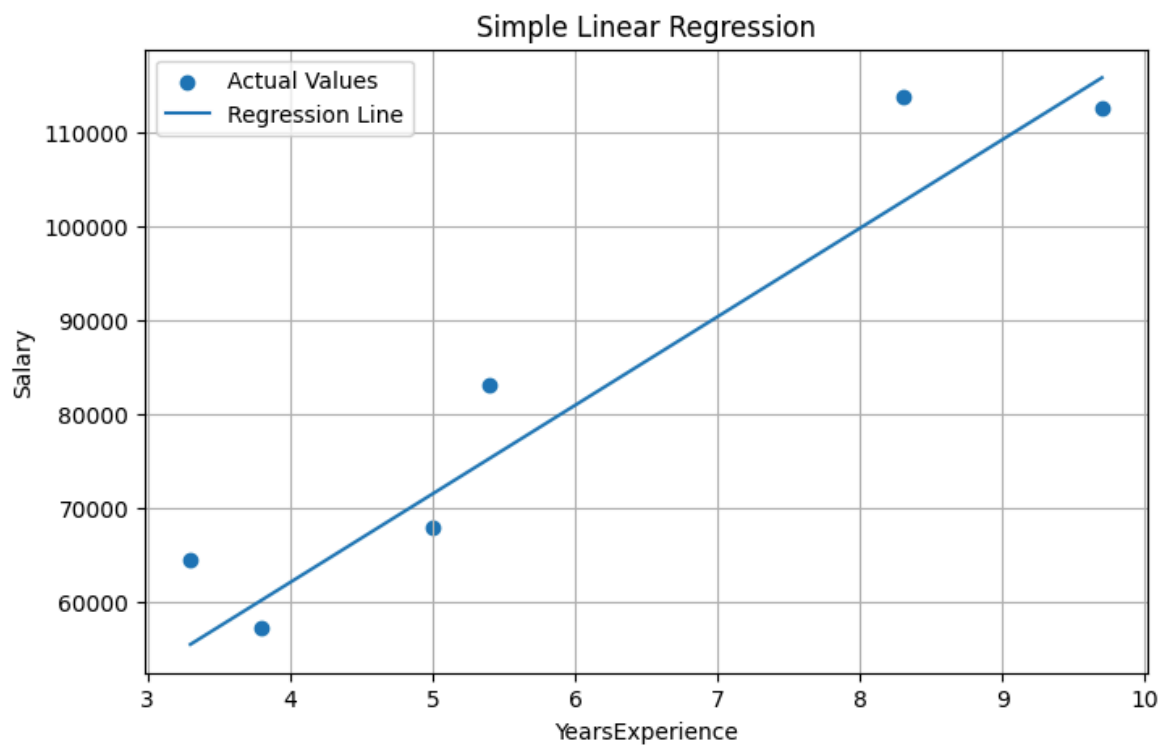
sorted_indices = np.argsort(X_test["YearsExperience"].values)

X_sorted = X_test["YearsExperience"].values[sorted_indices]
y_pred_sorted = y_predict[sorted_indices]

plt.plot(
    X_sorted,
    y_pred_sorted,
    label="Regression Line"
)

plt.title("Simple Linear Regression")
plt.xlabel("YearsExperience")
plt.ylabel("Salary")
plt.legend()
plt.grid(True)

plt.show()
```



In []: